

Problem 1. Multiply the following matrices.

1. $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$

2. $\begin{bmatrix} 9 & 6 \\ 3 & 2 \end{bmatrix} \begin{bmatrix} 9 & 6 \\ 3 & 2 \end{bmatrix}$

3. $\begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}^3$

Problem 2. Using the equations below for $\hat{\beta}_1$ and $\hat{\beta}_0$, argue that in the case of simple linear regression, the least squares line always pass through the point $(\bar{x}, \bar{y})$.

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

where $\bar{x}$ and $\bar{y}$ are the empirical means.

Problem 3. For the following exercise, consider the following linear model:

$$Y = \beta_0 + \beta_1 X + \epsilon$$

where ϵ is a Gaussian random variable following a normal distribution $N(0, \sigma^2)$. Using this model, we obtain the following training dataset:

$$\mathcal{D} = \{(x_i, y_i)\}_{i=1}^3 = \{(-1, -2), (0, 0), (1, 5)\}.$$

1. Find the coefficients for the model, $\hat{\beta}_0$ and $\hat{\beta}_1$.
2. Compute the RSS of the linear model.
3. Compute the RSE of the linear model.
4. Compute the TSS of the linear model.
5. Compute the R^2 of the linear model.

Problem 4. Consider a dataset with $n = 100$ points containing a single predictor ($p = 1$) and a quantitative response Y . To this dataset, we fit a linear regression model $\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 X$, as well as a separate cubic regression, i.e. $\hat{Y}' = \hat{\beta}'_0 + \hat{\beta}'_1 X + \hat{\beta}'_2 X^2 + \hat{\beta}'_3 X^3$.

1. Assume that nature is using a linear model to generate the dataset. Would you expect the training RSS of the linear regression to be smaller or larger than the training RSS of the cubic regression? Justify your answer.
2. Answer (1) using test RSS instead of training RSS.

3. Assume that nature is not using a linear model to generate the data, but we don't know how far it is from linear. Consider the training RSS for the linear regression, and also the training RSS for the cubic regression. Would we expect one to be lower than the other, would we expect them to be the same, or is there not enough information to tell? Justify your answer.
4. Answer **(3)** using test RSS instead of training RSS.

Problem 5. Assume that ‘nature’ behaves according to the following linear additive model: $Y = \beta_0 + \beta_1 X + \varepsilon$. After collecting a dataset $\mathcal{D}$ from nature, we would like to analyze the following hypothesis:

$$H_1: \beta_1 = 1 \text{ and } H_2: \beta_1 = 2.$$

Note: $SE(\hat{\beta}_1)$ is the same as $SD(\hat{\beta}_1)$.

1. Assuming that you compute $\hat{\beta}_1 = -3$ and $SE(\hat{\beta}_1) = 1$, what conclusions can you draw about the above hypothesis?
2. Assuming that you compute $\hat{\beta}_1 = 1/2$ and $SE(\hat{\beta}_1) = 2$, what conclusions can you draw about the above hypothesis?
3. Assuming that you compute $\hat{\beta}_1 = 2$ and $SE(\hat{\beta}_1) = 1/2$, what conclusions can you draw about the above hypothesis?

Problem 6. We would like to learn the parameters of a simple quadratic predictor of the form $\hat{Y} = \alpha X^2 + \gamma$. In particular, find the value of α that minimizes the RSS given the following dataset:

$$\mathcal{D} = \{(x_i, y_i)\}_{i=1}^3 = \left\{ (0, 0), \left(1, \frac{1}{2}\right), \left(-1, \frac{3}{2}\right) \right\}.$$